

Leftover Hash Lemma Extraction Bound for Unpredictable Random-Oracle Sources, Public Seed, k -Source Generalization

Status. Statement: AI-written, not yet formalized.

Category. Information-Theoretic Cryptography.

Placeholder. This is a lattice test entry, not yet reviewed for provenance: it asks whether joint multi-source extraction beats the naive k -fold hybrid bound obtained by repeating the single-source ($k = 1$) argument of crypto-conjura.github.io/c/0004 k times.

1 Setting

As in the $k = 1$ base case: fix nonempty finite $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. Take k independent ϵ -unpredictable sources $S_1, \dots, S_k$ over the same $(\mathcal{K}, \mathcal{D}, \mathcal{R})$, each producing its own (x_i, z_i) . Draw k independent seeds $sd_1, \dots, sd_k \xleftarrow{\$} \mathcal{K}$ and reveal all of them publicly, alongside all of H and all the z_i .

Definition 1 (k -Instance Public-Seed Extraction). For a challenge bit b , set $y_{i,0} \leftarrow H(sd_i, x_i)$ and $y_{i,1} \xleftarrow{\$} \mathcal{R}$ for each i , and let D output a guess b' given H , all sd_i , all $y_{i,b}$, and all z_i . Define

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub-}k}(S_1, \dots, S_k, D) := 2 \Pr[b' = b] - 1.$$

2 The Placeholder Conjecture

Proposition 1 (Trivial hybrid bound, not in question). *By a standard hybrid argument over the k instances, using the corollary of the $k = 1$ base case at each step,*

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub-}k}(S_1, \dots, S_k, D) \leq k \cdot \frac{8}{5} \left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}} \right).$$

At $k = 1$ this is exactly the base case's corollary, which is why that statement specializes this one.

Conjecture 1 (Placeholder, untested). *The linear-in- k factor above is not necessary: there is a bound of the same shape with the k -dependence replaced by $O(\sqrt{k})$ or better, exploiting that a single distinguisher must succeed jointly across all k instances rather than being handed k independent certificates.*

3 Open Obligations

- Decide whether this is worth pursuing as a real statement (drop the test category and attach real provenance) or leave as a lattice placeholder.
- If pursued: attempt to beat the trivial k -fold hybrid bound, or show it is tight.